Registers and Counters

Section A: Registers

Q1. Multi-function 4-bit register

Design a 4-bit register $Q_3Q_2Q_1Q_0$ using D flip-flops and one 8:1 multiplexer per bit. The mode is selected by $S_2S_1S_0$ as shown below. For the increment and carry modes, let $Cin = 0$. Ci denotes the carry generated by stage i (half-adder chain).

$S_2S_1S_0$	Operation (next-state input D_i)
000	Set to 0: $D_i = 0$
001	Set to 1: $D_i = 1$
010	Complement the existing value: $D_i = \overline{Q_i}$
011	Add 1 (input is the *sum*): $D_i = Q_i \oplus C_i$
100	Carry mode (input is the *carry of the previous stage*): $D_0 = Cin$ (initial carry), $D_{i+1} = C_i$
101	Shift right (serial input SI)
110	Shift left (serial input SI)
111	Hold: $D_i = Q_i$

(a) Write the Boolean expressions for the multiplexer inputs of every bit $D_0, \ldots, D_3$.

(b) Draw the logic diagram of the complete register.

(c) If $Q = 0110$, give the contents of the register after one clock pulse for each of the eight modes.

Q2. Shift register with subtractor and comparator

A serial bit stream X arrives MSB-first, one bit per clock. A 4-bit serial-in parallel-out (SIPO) shift register holds the present 4-bit word $A = Q_3Q_2Q_1Q_0$ (on every clock the newest bit enters Q_0 and the oldest bit is shifted out of Q_3). A 4-bit reference word $R = 0011$ is hardwired. All arithmetic is 4-bit (ignore the final borrow).

(a) Using a 4-bit subtractor, compute $D = A - R$ and then take the bitwise complement $\overline{D}$ of the result. Draw the block diagram.

(b) Using a magnitude comparator, extend the circuit so that the output $Z = 1$ if the complemented result $\overline{D}$ is greater than 10 (decimal), otherwise $Z = 0$.

(c) For the input stream $X = 0\,1\,0\,1\,0\,1\,0\,1\,1\,0\,1\,0$ (initial register contents 0000), tabulate A, D, $\overline{D}$ and Z after each clock pulse.

Q3. Shift register with XOR feedback

A 4-bit right-shift register shifts data as $Q_0 \to Q_1 \to Q_2 \to Q_3$. The serial input of Q_0 is driven by $D_0 = Q_2 \oplus Q_3$.

(a) Starting from $Q_0Q_1Q_2Q_3 = 1000$, write the register contents after each of the first six clock pulses.

(b) After how many clock pulses does the sequence first repeat? How many of the 16 possible states are visited in one cycle?

(c) To what value should the register be initialised so that the pattern 1000 appears after exactly two clock pulses?

Q4. Types of shift registers and timing

(a) List the four basic types of shift register (by input/output mode) and give one application of each.

(b) Design a system using a SIPO shift register that compares the present state of the serial input with its previous state and stores the result of the comparison in a 4-bit serial-in serial-out (SISO) register (result = 1 if the present and previous bits are equal, 0 otherwise). Draw the block diagram and, for the input stream $X = 1\,1\,0\,1\,0\,0\,1\,1$ (all registers initially 0), show the contents of the SIPO and SISO registers after each clock pulse.

(c) An 8-bit data word is to be sent over a serial link driven by a 2 MHz clock using a PISO register. How long does it take to transmit the word serially? How long would a parallel transfer take with a PIPO register on the same clock?

Section B: Counters

Q5. Up/Down BCD counter using T flip-flops

Design a synchronous 4-bit **up/down BCD (decade) counter** using T flip-flops. A control input $U/\overline{D}$ selects the direction: when $U/\overline{D} = 1$ the counter counts up $0 \to 1 \to \cdots \to 9 \to 0$, and when $U/\overline{D} = 0$ it counts down $9 \to 8 \to \cdots \to 0 \to 9$.

(a) Draw the state diagram and prepare the state table (present state, $U/\overline{D}$, next state, flip-flop inputs T_3, T_2, T_1, T_0).

(b) Using K-maps (treat states 10–15 as don't-cares), derive simplified expressions for T_3, T_2, T_1, T_0.

(c) Draw the logic diagram.

Q6. Synchronous vs. asynchronous reset

(a) Distinguish between a *synchronous* reset and an *asynchronous* reset of a flip-flop.

(b) A 4-bit binary up counter (with active-low asynchronous clear $\overline{CLR}$) is to be converted into a mod-10 counter. Draw the circuit using a NAND gate that detects the appropriate state and drives $\overline{CLR}$. Which state is detected, and how long does the counter actually stay in that state?

(c) Design the mod-10 counter using a 4-bit binary counter with a *synchronous* clear. Which state must now be detected, and why is it different from part (b)?

(d) Compare the two designs with respect to glitches on the outputs and the number of valid states displayed.

Q7. Mod-6 (3-bit, six-state) counter

Design a synchronous 3-bit counter that has six stages and counts $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 0$ using D flip-flops.

(a) Draw the state table and obtain the expressions for D_2, D_1, D_0 using K-maps.

(b) Draw the logic diagram.

(c) What happens if the counter goes into an unused state 6 or 7? Is the design self-correcting? Show on the state diagram.

Q8. Ripple vs. synchronous counter

(a) State two differences between a ripple (asynchronous) counter and a synchronous counter.

(b) How many flip-flops are needed to count from 0 to $N-1$? Evaluate for $N = 60$.

(c) Each flip-flop has a clock-to-Q delay of 10 ns and each AND gate has a delay of 5 ns. Using the 4-bit binary ripple counter and the 4-bit synchronous binary counter designs given in Morris Mano's *Digital Design*, find the time required for the output to settle to its next state, and hence the maximum clock frequency, for (i) the ripple counter and (ii) the synchronous counter.

Q9. Ring and Johnson counters

(a) Draw a 4-bit ring counter (initial state 1000) and a 4-bit Johnson (twisted-ring) counter (initial state 0000). Write the state sequence of each.

(b) State the modulus of an N-flip-flop ring counter and of an N-flip-flop Johnson counter. How many unused states does each have?

(c) List the unused states of a 3-bit Johnson counter.

(d) How many flip-flops are required to build a mod-12 Johnson counter? How many 2-input AND gates are needed to decode all 12 states?

Q10. Frequency division

(a) A 4-bit binary counter is clocked at 160 MHz. Find the frequency of the outputs Q_0, Q_1, Q_2, Q_3 and the duty cycle of Q_3.

(b) A 10 kHz clock must be divided down to 1 Hz. Show how this can be done using cascaded decade (mod-10) counters. How many counters and how many flip-flops are required?

(c) Design a divide-by-3 circuit ($f_{out} = f_{in}/3$) using D flip-flops that produces the output with (i) a 66.67% duty cycle and (ii) a 50% duty cycle. Draw the circuits and the timing diagrams of the clock and the output. (Hint: the 50% case needs both clock edges.)